during this period, both the wire and the condenser, C, connected with the wire, are at a certain voltage which can be amplified by methods commonly employed in broadcasting technique. As is customarily done in most of these measurements, a counting device, similar to a telephone counter, can be attached, or the voltage can be transferred to a loudspeaker, and thus one can count and register every particle that passes through the counting tube.

The number of the ions produced by beta and gamma radiation is small. Therefore, big amplification is required, and it is customary to work in the resolving region. Furthermore, since the beta particles are not very penetrating, thin-walled tubes are used for counting beta particles; on the other hand, for gamma-ray photons thick-walled tubes are used, so as to keep other types of radiation out as much as possible. When dealing with alpha rays, which produce far more electrons, big amplification can be dispensed with, and it is possible to work in the proportional region. The advantage of this method is that, if the counting device is properly connected, it does not react to other types of radiation. The latter produce weak potential impulses only. A special amplifier, called the *thyratron,* which transmits impulses above a certain wavelength only, permits the weaker impulses to be eliminated, so that only those produced by alpha radiation are counted. This is important mainly because, in addition to the radiation under investigation, all other conceivable types of penetrating radiation are roaming through space. In the first place, electrons are liberated everywhere, even in the counter itself, by cosmic radiation which cannot be screened sufficiently by any known means. Secondly, no existing substance is entirely free from radioactive impurities, so that even the material of which the counter is made tends to release impulses occasionally. Such limited effects are simply just inevitable with these measuring devices. When counting alpha particles, the counting tube must be equipped with a thin window of mica, to enable the particles to enter, since they would not be able to pass through anything thicker.

Another very important instrument of the nuclear physicist is Wilson's *cloud chamber,* the operation of which was explained in our second lecture. The advantage of this device consists principally in the fact that it permits us to obtain a visual record of nuclear processes, thus showing simultaneously a great many of the details of the process.

Figure 32 shows a simple sketch of the cloud chamber. The upper section contains air, saturated with water vapour. It is covered with a glass plate at the top to permit observation; below there is a movable piston, covered by a damp layer of gelatine, so that the air above it is kept saturated with water vapour. The light necessary for observing the cloud tracks is admitted through an aperture in the side. The piston is suddenly moved downward, so that the air expands adiabatically and cools. The result is that the water vapour becomes supersaturated and the ionization produced by a particle entering the